# Cut a table, lose the answer

Day 03: chunking. Same table, same retriever, two ways to cut.

**Q: What is the hotel limit per night in Singapore?**

### A · CUT EVERY 40 WORDS

`travel-policy#w080`

... reimbursed up to the daily limits ...

| City | Hotel per night | Meals per day |
|---|---|---|
| London | USD 250 | USD 75 |
| New York | USD | |

`travel-policy#w120`

| | | |
|---|---|---|
| 300 | USD 75 | |
| Singapore | USD 220 | USD 60 |
| Other cities | USD 180 | USD 50 |

220 or 60? No column header.

### B · CUT AT HEADINGS

`travel-policy#daily-limits-by-city`

**Travel Policy > Daily limits by city**

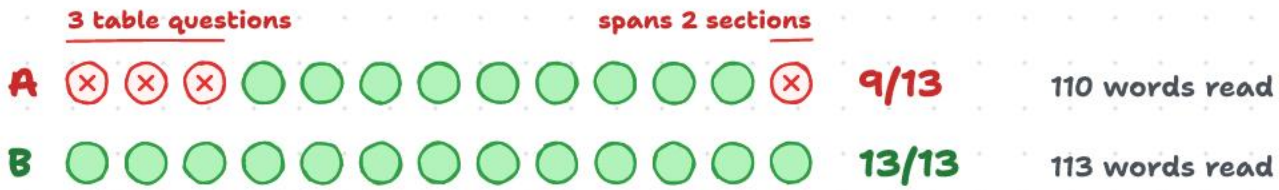
Hotel and meal costs are reimbursed ...

| City | Hotel per night | Meals per day |
|---|---|---|
| London | USD 250 | USD 75 |
| New York | USD 300 | USD 75 |
| Singapore | USD 220 | USD 60 |
| Other cities | USD 180 | USD 50 |

USD 220 per night. One chunk.

## 13 test questions · Day 01's BM25, top 3

Is the complete evidence inside one retrieved chunk?




**Retrieval found the table both times. Only one chunk could answer.**

Real run · Day 03 notebook · free, no API key

# Cut along the structure

Four rules. They work on Markdown, HTML and PDF alike.

1 H **SPLIT AT HEADINGS**

## starts a section · text before it becomes "Overview"

2 **KEEP UNITS WHOLE**

paragraphs, list items, whole tables · never cut mid-row

3 **PACK UP TO A LIMIT**

80 words here · too long? split between units

4 **STAMP EVERY CHUNK**

Title > Section · stable ID · groups · hashes

## TABLE TOO LONG? SPLIT BETWEEN ROWS, REPEAT THE HEADER ROW

real output, limit 30 words

…#daily-limits-by-city-part-3

**Travel Policy > Daily limits by city**

| City | Hotel per night | Meals per day |
|---|---|---|
| London | USD 250 | USD 75 |
| New York | USD 300 | USD 75 |
| Singapore | USD 220 | USD 60 |

…#daily-limits-by-city-part-4

**Travel Policy > Daily limits by city**

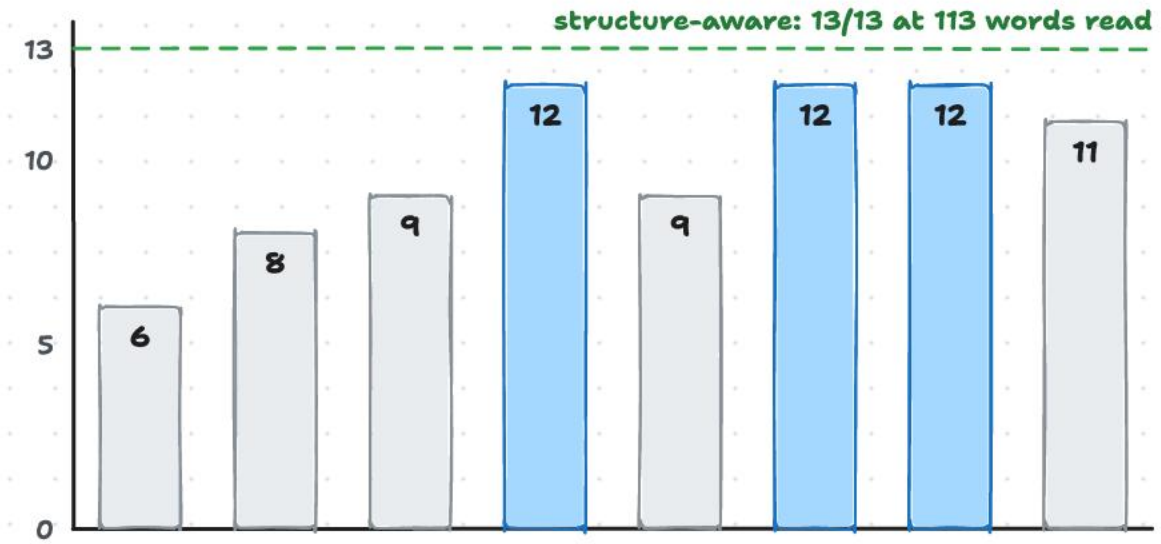
| City | Hotel per night | Meals per day |
|---|---|---|
| Other cities | USD 180 | USD 50 |

**Without the repeated header, "USD 180" would mean nothing.**

Parts 1–2: the paragraph above the table, split between sentences.

# Bigger isn't steadily better

Fixed windows, no overlap: complete evidence in one chunk, of 13.




| window | 20 | 30 | 40 | 50 | 60 | 80 | 100 | 120 |
|---|---|---|---|---|---|---|---|---|
| table kept | ✗ | ✗ | ✗ | ✓ | ✗ | ✓ | ✓ | ✗ |
| words read | 59 | 87 | 110 | 139 | 166 | 194 | 242 | 284 |

Blue bars and ✓: the Singapore row stays in one chunk with its column header.

### OVERLAP (40 + 10 words)

- Repaired 3 of 4 misses
- +27% text to embed and store
- Software table still headerless

### "TITLE > SECTION" HEADER

- Helped 2 questions
- Hurt 1: parental leave 2 → 3
- +18 words read per question

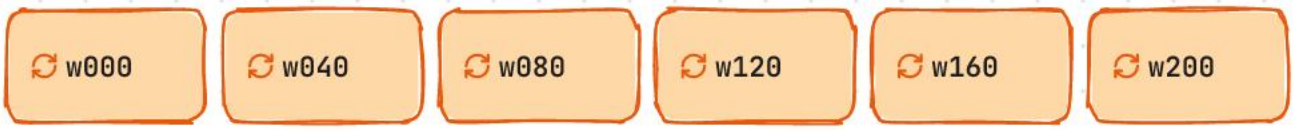
**Fixed cuts land by luck. Measure the knobs; don't assume.**

# One edit: how many chunks change?

Add 7 words to the top of travel-policy, then re-chunk.

## A · FIXED 40 WORDS

6 of 6 changed




the 7 new words push every boundary along

Re-embed all six. [travel-policy#w080] now cites different text.

## B · STRUCTURE-AWARE

1 of 5 changed




Only the edited section needs re-embedding.

Section IDs such as travel-policy#daily-limits-by-city keep their meaning.

## EVERY CHUNK CARRIES ITS DOCUMENT'S METADATA

| | |
|---|---|
| chunk_id | travel-policy#daily-limits-by-city |
| allowed_groups | all-employees |
| doc_hash | bf4bdf9b29ce… |
| chunk_hash | e13a0fb0b50f… |

Manager pay guide: 1 chunk, groups managers + people-team.

Filters need this on every chunk (Day 05).

Recording permissions is not enforcing them: that is Day 14.

**Next → Day 04: embeddings and similarity**